

Podcast 27 — Migration Toolkit

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Length target: 15 min **Repo references:** `domain-4-migration/01-cloud-migrations.md`, `02-dms.md`, `03-mgn.md`, `04-datasync.md`, `05-snow-family.md`, `06-storage-gateway.md`, `domain-2-new-solutions/37-other-services.md` (Transfer Family)

Topic & Scope

The 7 R's, migration discovery + planning tools, server migration (MGN), database migration (DMS + SCT), data transfer (DataSync, Snow Family, Transfer Family, Storage Gateway).

Service Coverage Depth

Deep: - 7 R's (Retire, Retain, Rehost, Relocate, Repurchase, Replatform, Refactor) - AWS Application Migration Service (MGN) — block-level for servers - AWS DMS + Schema Conversion Tool (SCT) — database migration - DataSync — file-level network migration

Brief: - Snow Family (Snowcone, Snowball Edge, Snowmobile) - Transfer Family (SFTP / FTPS / FTP / AS2) - Migration Hub + Application Discovery Service - Migration Evaluator

Structured Outline

1. **Open (30s)** — “Migration is Domain 4 — 20% of the exam. The 7 R's + a handful of services cover most questions.”
2. **The 7 R's (3 min)** — Retire (decommission), Retain (keep on-prem), Rehost (lift & shift), Relocate (move VMware to AWS Cloud), Repurchase (move to SaaS), Replatform (lift, tinker, shift), Refactor (re-architect)
3. **Discovery & Planning (2 min)** — Application Discovery Service (agent-based + agentless), Migration Hub (status tracking), Migration Evaluator (TCO + sizing)
4. **MGN (3 min)** — agent-based continuous block-level replication, source → staging → cutover, support for any OS, replaces SMS
5. **DMS + SCT (3.5 min)** — DMS migrates data (homogeneous + heterogeneous), CDC for ongoing replication, SCT converts schema (Oracle/SQL Server → PostgreSQL/MySQL/Aurora), DMS Serverless option
6. **DataSync + Snow Family + Transfer Family (2.5 min)** — DataSync (network bulk move), Snow Family (offline shipping when bandwidth limited), Transfer Family (managed SFTP/FTPS/FTP/AS2)
7. **Rapid-Fire Trap Drill (60s)**

Must-Mention Exam Tips

- 7 R's: memorize the names + a sentence definition each

- Rehost = lift-and-shift — fastest, least optimization
- Replatform = “lift, tinker, shift” — small changes (e.g., RDS instead of self-managed MySQL)
- Refactor = re-architect (containers, serverless, etc.)
- Application Discovery Service: agent (deeper data) + agentless (VMware-based)
- Migration Hub: dashboard across MGN / DMS / Discovery
- Migration Evaluator (formerly TSO Logic): TCO analysis + right-sizing for migration planning
- MGN: continuous block-level replication, RPO seconds, supports any OS that boots on EC2 nitro instance types
- MGN replaces CloudEndure Migration + Server Migration Service (SMS)
- DMS: homogeneous (MySQL → MySQL) + heterogeneous (Oracle → Aurora PG) migrations
- DMS CDC (Change Data Capture) keeps source + target in sync for cutover
- SCT: Schema Conversion Tool — converts schema between engines, generates code change reports
- DMS Serverless: auto-scales capacity, pay-as-you-go
- DataSync: agent on source, transfers to S3/EFS/FSx, 10× faster than rsync, automatic encryption + integrity checks
- DataSync sources: NFS, SMB, HDFS, Object Storage (S3-compatible), Google Cloud Storage, Azure Files
- Snow Family: Snowcone (small, edge, 8 TB), Snowball Edge Storage Optimized (80 TB usable), Snowball Edge Compute Optimized (with GPU option), Snowmobile (100 PB truck — being phased out)
- Snow Family encrypts at rest (256-bit) + in transit; e-ink shipping labels
- Transfer Family: managed SFTP / FTPS / FTP / AS2 backed by S3 or EFS, custom auth via Lambda, supports user lifecycle

Must-Mention Exam Traps

- EXAM TRAP: “Lift and shift” = REHOST; “lift, tinker, shift” = REPLATFORM — different
- EXAM TRAP: MGN ≠ DRS — MGN is for migration (one-time cutover); DRS is for ongoing DR
- EXAM TRAP: DMS migrates DATA, not schema — for schema use SCT
- EXAM TRAP: DMS CDC has source-engine support requirements — Oracle needs LogMiner / Binary Reader
- EXAM TRAP: DataSync is for online network transfer — for limited bandwidth or air-gapped use Snow Family
- EXAM TRAP: DataSync requires DataSync agent on source — except for AWS-to-AWS (no agent needed)
- EXAM TRAP: Snowball Edge has compute — can run EC2 + Lambda offline
- EXAM TRAP: Snowmobile (100 PB truck) is being phased out — most exam questions use Snowball Edge variants
- EXAM TRAP: Transfer Family is for incoming file transfers — NOT for AWS-to-on-prem (use DataSync or Storage Gateway)

- EXAM TRAP: Application Discovery + Migration Hub work together — Discovery feeds Hub
- EXAM TRAP: VMware Cloud on AWS is a Relocate strategy (move VMs with same hypervisor)
- EXAM TRAP: 7 R's pick depends on cost / complexity tradeoff — usually Replatform is “sweet spot” for SAP-C02
- EXAM TRAP: SCT generates a code conversion REPORT — manual rework still needed for stored procedures

Key Decision Matrix

Scenario	Pick
Move servers as-is to EC2	Application Migration Service (MGN)
Move database with engine change	DMS + SCT
Continuously replicate DB during migration	DMS CDC
Inventory + plan on-prem servers	Application Discovery Service
Track migration progress across services	Migration Hub
TCO analysis + sizing for migration	Migration Evaluator
Online file transfer at scale	DataSync
Offline transfer (PB+, limited bandwidth)	Snowball Edge
Small edge data collection (TB)	Snowcone
Hybrid on-prem file share via SMB/NFS backed by S3	Storage Gateway File Gateway
Inbound SFTP / FTP / AS2 to S3 / EFS	Transfer Family
Re-architect to containers / serverless	Refactor
Lift and shift simple workload	Rehost (MGN)
Modify slightly (managed DB swap)	Replatform

Tone & Style

- Sing-song the 7 R's with examples
- Frame MGN as “lift-and-shift” engine
- Repeat “DMS = data, SCT = schema”

Rapid-Fire Closer

“Lift-and-shift servers?” — “MGN (Rehost).” “DB engine change?” — “DMS + SCT.” “Limited bandwidth bulk move?” — “Snowball Edge.” “Online file transfer?” — “DataSync.” “Inbound SFTP?” — “Transfer Family.” “Migration progress dashboard?” — “Migration Hub.”